

Study Design and Rigor

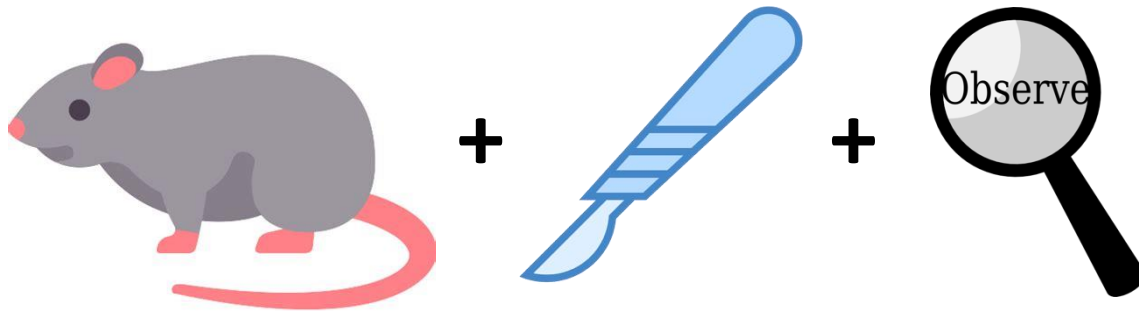
Timothy Houle, PhD

Massachusetts General Hospital

What We Talk About When We Talk About Rigor

- Animal pain models are exercises in measurement theory
 - Determine how well therapeutic responses to a laboratory model (measure) corresponds with clinical response (criterion)
- Good measurement involves
 - Validity
 - Reliability

Validity: Does the design measure what its supposed to measure?

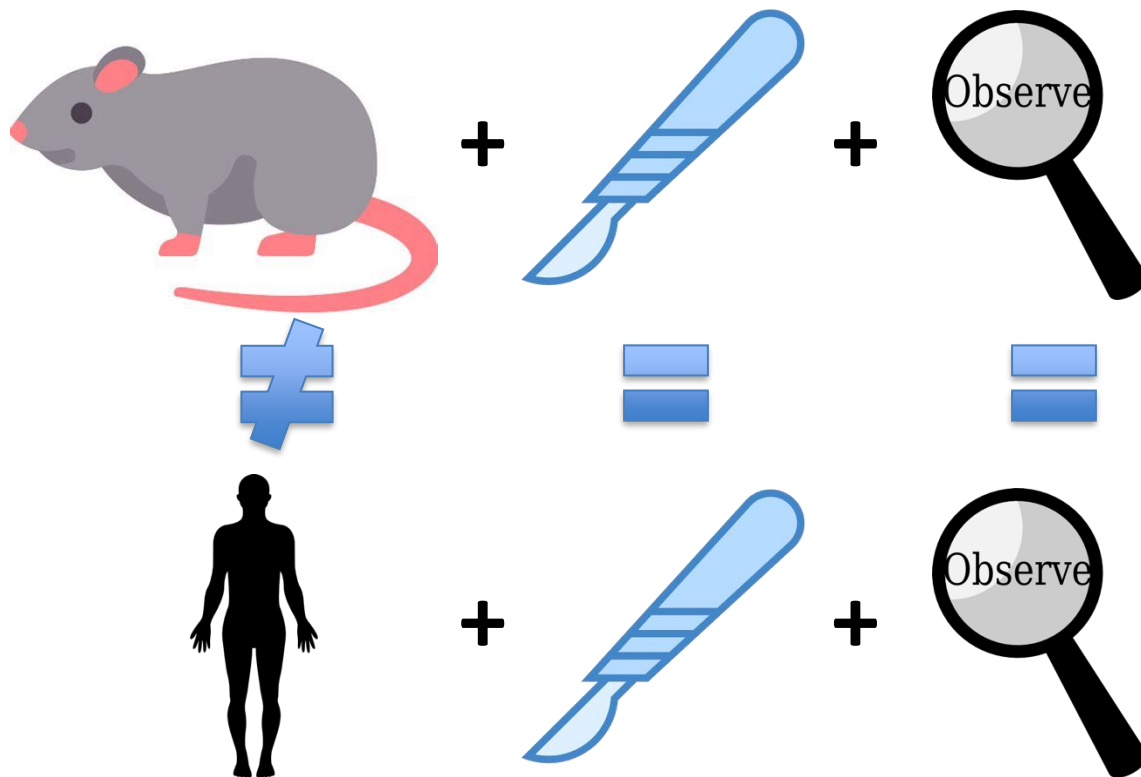


Animal

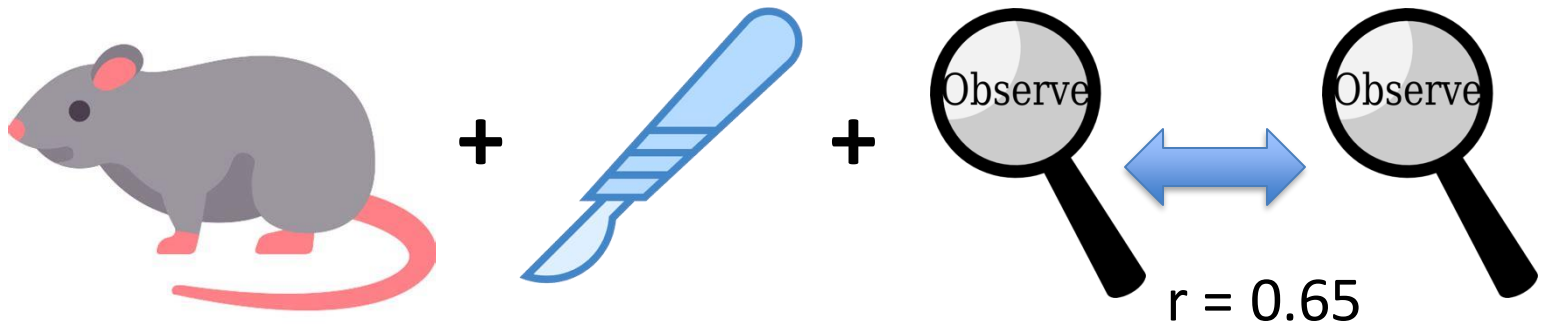
Insult/Challenge

Measure/Behavior

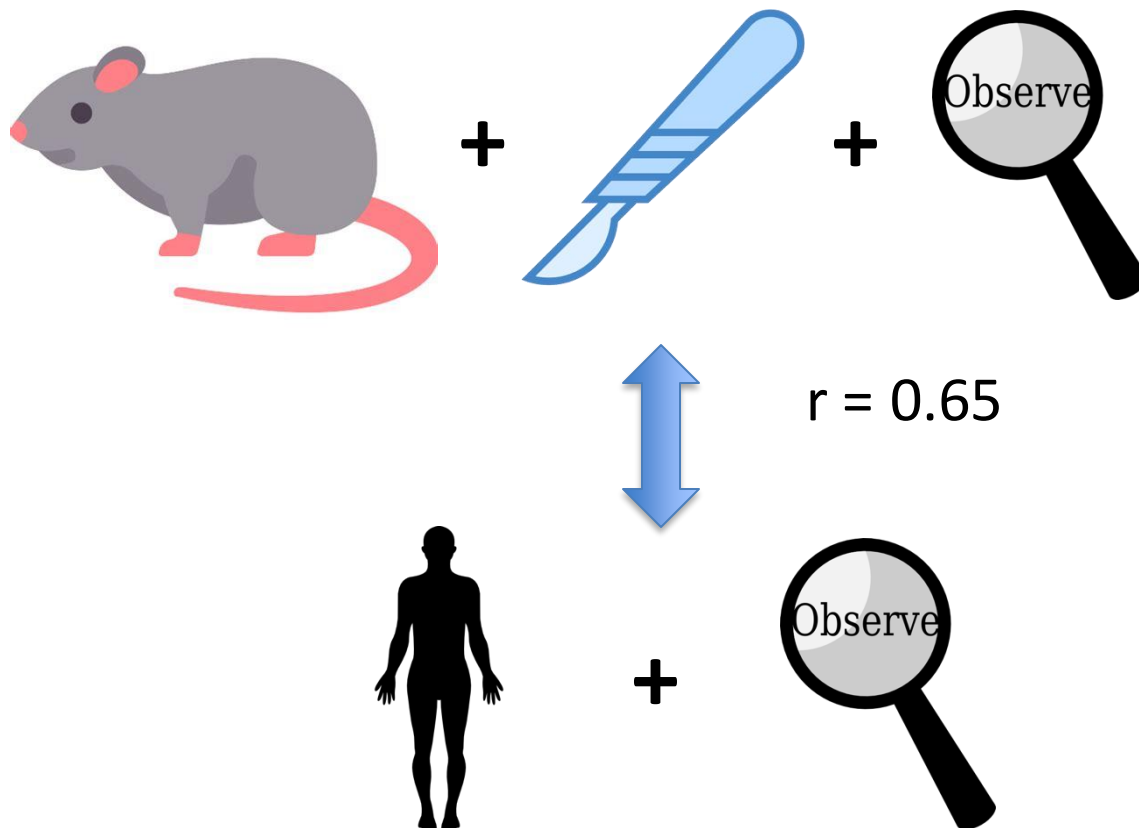
Content Validity: The extent to which the model covers the target domain



Concurrent Validity: The correlation between different model outputs measured simultaneously



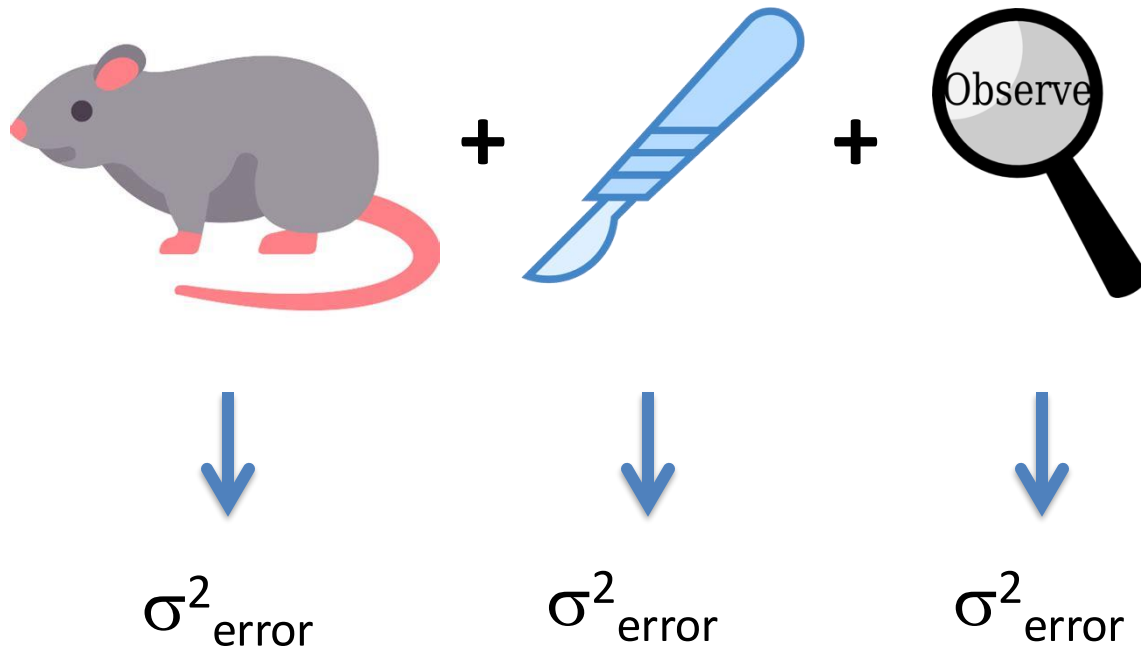
Predictive (Criterion) Validity: The association between the model and future human findings



What We (Should) Talk About When We Talk About Validity

- Content Validity: The extent to which the model covers the target domains
 - Animal
 - Insult
 - Measurement
- Concurrent Validity: Does the model offer different model outputs measured simultaneously for comparison?
- Predictive (Criterion) Validity: What is the expected association between the model and future human findings?

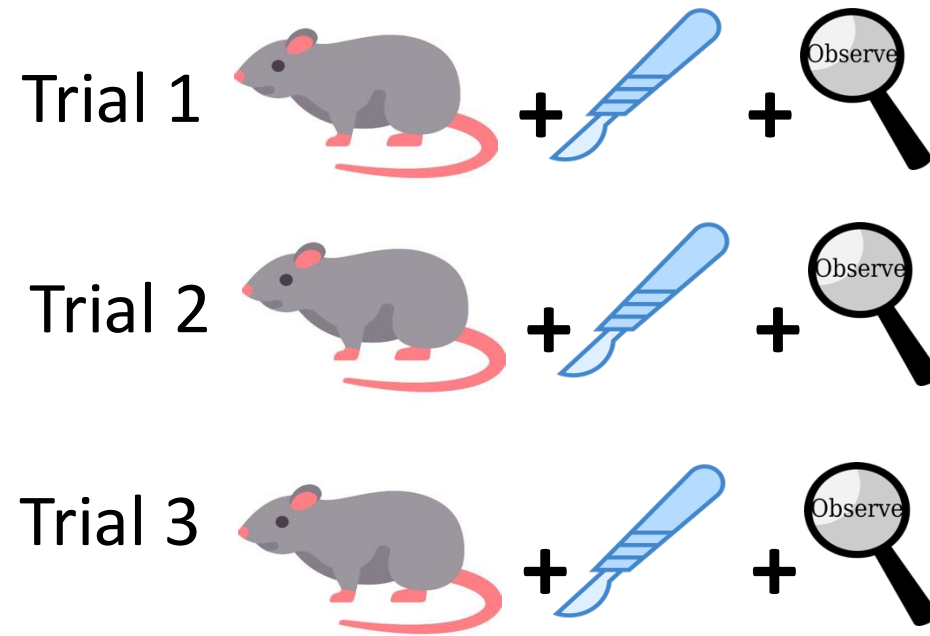
Reliability: What is the consistency of the design under similar conditions?



$$\rho_{xx} = \sigma^2_{\text{error}} / \sigma^2_{\text{total}}$$

Reliability: What is the consistency of the design under similar conditions?

Lab A



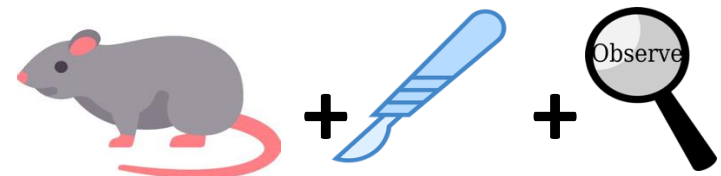
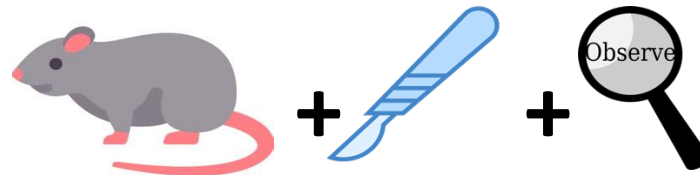
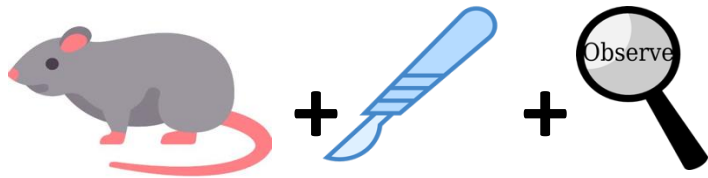
Average correlation = ρ_{xx}

Reliability: What is the consistency of the design under similar conditions?

Lab A

Lab B

Lab A



Average correlation = ρ_{xx}

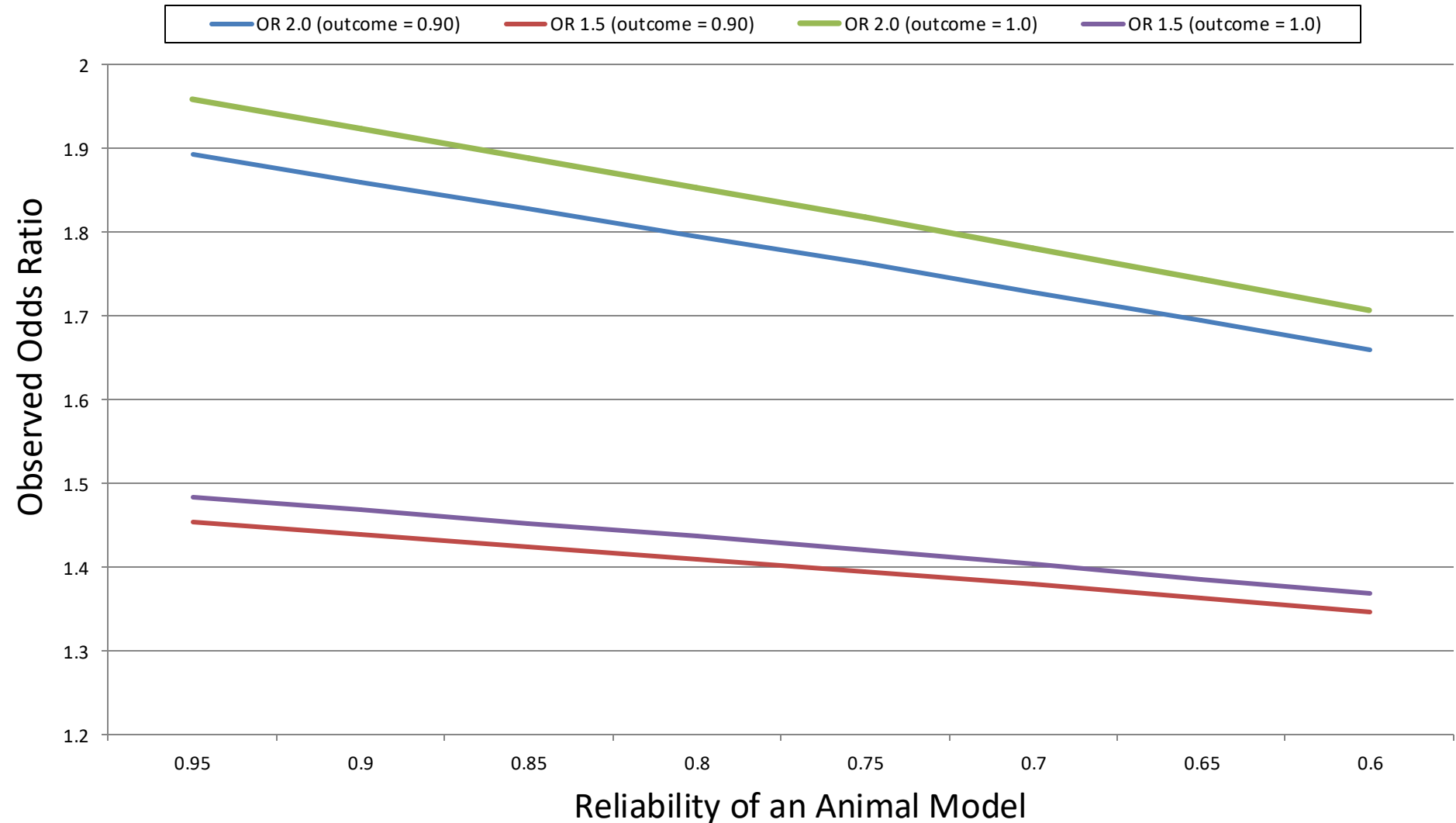
The reliability of an animal model

- Studies of animal models seldom explore the statistical reliability of the models
 - Perhaps because the marker is drawn from a biological sample, it is assumed to have high reliability (i.e., precision).
- However, the measurement from animal models can be obscured by error like any other measurement
 - Measurement error (random error)
 - Systematic error (bias)
- Reliability and validity are contextual (i.e., for a specific setting)

$$\text{"True" association}_{xy} = \frac{\text{Observed Association}_{xy}}{\text{Reliability}_x * \text{Reliability}_y} = r_{x'y'} = \frac{r_{xy}}{\sqrt{r_{xx} * r_{yy}}}$$

Association between animal study and human study

“True” odds ratio (reliability of outcome)



What We (Should) Talk About When We Talk About Reliability

- Theoretical reliability of the model: What is the consistency of the model under similar conditions?
- Sources of error (variability): Would certain design choices reduce error variance?
- Inter-lab reliability: what models or design choices would lead to cross-lab consistency?

Bias: When the estimated treatment effect differs from that in the population

- There is growing evidence that research findings are not simply unreliable, but biased.
- Bias has been attributed to many factors (e.g., researcher degrees of freedom, COI, etc.)
- However, design choices have recently been isolated to be related to statistical estimation bias

Garden of Forking Paths

Gelman (2013)

Scenario	Test Statistic	
1. Simple classical test	$T(y)$	One planned statistical inference
2. Test pre-chosen from set of possible tests	$T(y; \phi)$	One test with pre-registered ϕ
3. Test based on the data	$T(y; \phi(y))$	Only one test. Different test would have been performed given different data
4. Fishing	$T(y; \phi_j)$	Performing j tests and reporting the best one(s)

ϕ : control variables, covariates, transformations, data coding rules, exclusion, outliers, main effects, interactions, subgroups, alternate outcomes, direction of effect

Thank You

- Good Luck!
- `thoule1@mgh.harvard.edu`